Quantum mechanics pg 443-454
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Complement F_{IV}

SPIN 1/2 PARTICLE IN A STATIC MAGNETIC FIELD AND A ROTATING FIELD: MAGNETIC RESONANCE

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In chapter IV, we used quantum mechanics to study the evolution of a spin 1/2 in a static magnetic field. In this complement, we shall consider the case of a spin 1/2 simultaneously subjected to several magnetic fields, some of which can be time dependent, as is the case in magnetic resonance experiments. Before attacking this problem quantum mechanically, we shall briefly review several results obtained using classical mechanics.

1. Classical treatment ; rotating reference frame

a. MOTION IN A STATIC FIELD; LARMOR PRECESSION

Consider a system of angular momentum **j** which possesses a magnetic moment $\mathbf{m} = \gamma\mathbf{j}$ collinear with **j** (the constant γ is the gyromagnetic ratio of the system), placed in a static magnetic field $\mathbf{B}_0$, which exerts a torque $\boldsymbol{m} \times \mathbf{B}_0$ on the system. The classical equation of motion of **j** is :

$$\frac{d\mathbf{j}}{dt} = \mathbf{m} \times \mathbf{B}_0 \tag{1}$$

or:

$$\frac{d}{dt}\mathbf{m}(t) = \gamma\mathbf{m}(t) \times \mathbf{B}_0 \tag{2}$$

Performing a scalar multiplication of both sides of this equation by either **m**(t) or $\mathbf{B}_0$, we obtain :

$$\frac{d}{dt}[\mathbf{m}(t)]^2 = 0 \tag{3}$$

$$\frac{d}{dt}[m(t) \cdot \mathbf{B}_0] = 0 \tag{4}$$

m(t) therefore evolves with a constant modulus, maintaining a constant angle with $\mathbf{B}_0$. If we project equation (2) onto the plane perpendicular to $\mathbf{B}_0$, we see that m(t) rotates about $\mathbf{B}_0$ (Larmor precession) with an angular velocity of $\omega_0 = -\gamma B_0$ (the rotation is counterclockwise if γ is positive).

b. INFLUENCE OF A ROTATING FIELD; RESONANCE

Now assume that we add to the static field $\mathbf{B}_0$ a field $B_1(t)$, perpendicular to $\mathbf{B}_0$, and which is of constant modulus and rotates about $\mathbf{B}_0$ with an angular velocity ω (*cf.* fig. 1). We set:

$$\omega_0 = -\gamma B_0$$
$$\omega_1 = -\gamma B_1 \tag{5}$$

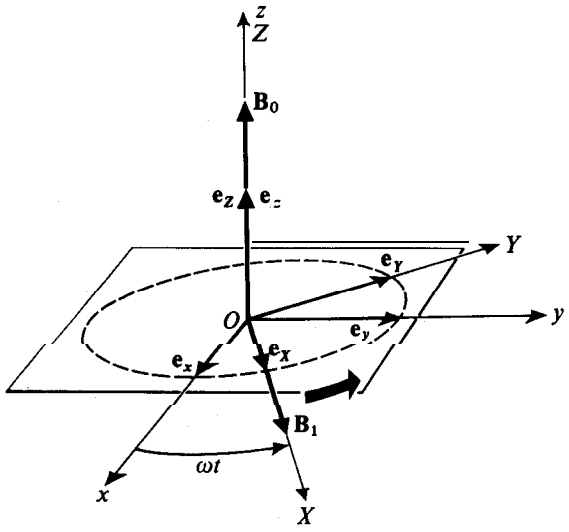


FIGURE 1

Oxyz is a fixed coordinate system. The static magnetic field $\mathbf{B}_0$ is directed along the *Oz* axis. The *OXYZ* system [the *OX* axis is the direction of the field $B_1(t)$] rotates about *Oz* with the angular velocity w.

We shall designate by *Oxyz* (unit vectors e_x, $\mathbf{e}_y$, e_z) a fixed coordinate system, whose *Oz* axis is the direction of the field B_o, and by *OXYZ* (unit vectors $\mathbf{e}_X$, $\mathbf{e}_Y$, $\mathbf{e}_Z$), the axes obtained from *Oxyz* by rotation through an angle ωt about *Oz* [*OX* is the direction of the rotating field $B_1(t)$]. The equation of motion of m(t) in the presence of the total field $B(t) = \mathbf{B}_0 + B_1(t)$ then becomes :

$$\frac{d}{dt}\mathbf{m}(t) = \gamma\, \mathbf{m}(t) \times [\mathbf{B}_0 + \mathbf{B}_1(t)] \tag{6}$$

To solve this equation, it is convenient to place ourselves, not in the absolute reference frame *Oxyz*, but in the rotating reference frame *OXYZ*, with respect to which the relative velocity of the vector m(t) is:

$$\left(\frac{d\mathbf{m}}{dt}\right)_{rel} = \frac{dm}{dt} - \omega\, \mathbf{e}_Z \times \mathbf{m}(r) \tag{7}$$

Let us set :

$$\Delta\omega = \omega - \omega_0 \tag{8}$$

Substituting (6) into (7), we obtain:

$$\left(\frac{d\mathbf{m}}{dt}\right)_{rel} = \mathbf{m(t)} \times [\Delta\omega\ \mathbf{e}_Z - \omega_1 \mathbf{e}_X] \tag{9}$$

This equation is much simpler to solve than equation (6), since the coefficients of the right-hand side are now time-independent. Moreover, its form is analogous to that of (2) : the relative motion of the vector m(t) is therefore a rotation about the "effective field" $\mathbf{B}_{eff}$ (which is static with respect to the rotating reference frame), given by (*cf.* fig. 2):

$$\mathbf{B}_{eff} = \frac{1}{\gamma}[\Delta\omega\ \mathbf{e}_Z - \omega_1 \mathbf{e}_X] \tag{10}$$

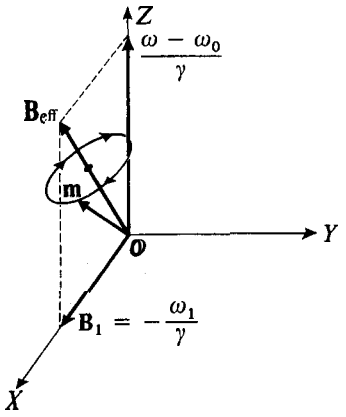


FIGURE 2

In the rotating reference frame *OXYZ*, the effective field $\mathbf{B}_{eff}$ has a fixed direction, about which the magnetic moment m(t) rotates with a constant angular velocity (precession in the rotating reference frame).

To obtain the absolute motion of m(t), we must combine this precession about $\mathbf{B}_{eff}$ with a rotation about Oz of angular velocity ω.

These first results already enable us to understand the essence of the magnetic resonance phenomenon. Let us consider a magnetic moment which, at time $t = 0$, is parallel to the field $\mathbf{B}_0$ (the case, for example, of a magnetic moment in thermodynamic equilibrium at very low temperatures : it is in the lowest energy state possible in the presence of the field $\mathbf{B}_0$). What happens when we apply a weak rotating field $B_1(t)$? If the rotation frequency $\omega/2\pi$ of this field is very different from the natural frequency $\omega_0/2\pi$ (more precisely, if $\Delta\omega = \omega - \omega_0$ is much larger than ω_1), the effective field is directed practically along Oz. The precession of m(r) about $\mathbf{B}_{eff}$ then has a very small amplitude and hardly modifies the direction of the magnetic moment. On the other hand, if the resonance condition $\omega \simeq \omega_0$ is satisfied ($\Delta\omega \ll \omega_1$), the angle between the field $\mathbf{B}_{eff}$ and *Oz* is large. The precession of the magnetic moment then has a large amplitude and, at resonance ($\Delta\omega = 0$), the magnetic moment can even be completely flipped.

2. Quantum mechanical treatment

a. THE SCHRODINGEr EQUATION

Let $|+\rangle$ and $|-\rangle$ be two eigenvectors of the projection S_z of the spin onto Oz, with respective eigenvalues + h/2 and – h/2. The state vector of the system can be written:

$$|\psi(t)\rangle = a_+(t)\,|+\rangle + a_-(t)\,|-\rangle \tag{11}$$

The Hamiltonian operator H(t) of the system is*:

$$H(r) = -M \cdot B(t) = -\gamma S \cdot [B_0 + B_1(t)] \tag{12}$$

that is, expanding the scalar product :

$$H(t) = \omega_0 S_z + \omega_1[\cos \omega t\, S_x + \sin \omega t\, S_y] \tag{13}$$

Using formulas (A-16) and (A-17) of chapter IV, we obtain the matrix which represents H in the $\{|+\rangle, |-\rangle\}$ basis :

$$H = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix} \tag{14}$$

Using (11) and (14), we can write the Schrodinger equation in the form:

$$\begin{cases} i\dfrac{d}{dt} a_+(t) = \dfrac{\omega_0}{2} a_+(t) + \dfrac{\omega_1}{2} e^{-i\omega t} a_-(t) \\ i\dfrac{d}{dt} a_-(t) = \dfrac{\omega_1}{2} e^{i\omega t} a_+(t) - \dfrac{\omega_0}{2} a_-(t) \end{cases} \tag{15}$$

b. CHANGING TO THE ROTATING FRAME

Equations (15) constitute a linear homogeneous system with time-dependent coefficients. It is convenient to define new functions by setting :

$$\begin{aligned} b_+(t) &= e^{i\omega t/2} a_+(t) \\ b_-(t) &= e^{-i\omega t/2} a_-(t) \end{aligned} \tag{16}$$

Substituting (16) into (15), we obtain a system which has constant coefficients :

$$\begin{cases} i\dfrac{d}{dt} b_+(t) = -\dfrac{\Delta\omega}{2} b_+(t) + \dfrac{\omega_1}{2} b_-(t) \\ i\dfrac{d}{dt} b_-(t) = \dfrac{\omega_1}{2} b_+(t) + \dfrac{\Delta\omega}{2} b_-(t) \end{cases} \tag{17}$$

* In expression (12), M . B(t) symbolizes the scalar product $M_x B_x(t) + M_y B_y(t) + M_z B_z(t)$, where M_x, M_y and M_z are operators (observables of the system under study), while $B_x(t)$, $B_y(t)$ and $B_z(t)$ are numbers (since we consider the magnetic field to be a classical quantity whose value is imposed by an external device which is independent of the system under study).

This system can also be written :

$$i\hbar \frac{d}{dt} \left| \tilde{\psi}(t) \right\rangle = \tilde{H} \left| \tilde{\psi}(t) \right\rangle \tag{18}$$

if we introduce the ket $|\tilde{\psi}(t)\rangle$ and the operator $\tilde{H}$ defined by :

$$|\tilde{\psi}(t)\rangle = b_+(t)|+\rangle + b_-(t)|-\rangle \tag{19}$$

$$\tilde{H} = \frac{\hbar}{2}\begin{pmatrix} -\Delta\omega & \omega_1 \\ \omega_1 & \Delta\omega \end{pmatrix} \tag{20}$$

Transformation (16) has led to equation (18), which is analogous to a Schrodinger equation in which the operator $\tilde{H}$, given in (20), plays the role of a time-independent Hamiltonian. $\tilde{H}$ describes the interaction of the spin with a *fixed* field, whose components are none other than those of the effective field introduced above in the $OXYZ$ frame [formula (10)]. We can therefore consider that the transformation (16) is the quantum mechanical equivalent of the change from the fixed *Oxyz* frame to the rotating $OXYZ$ frame.

This result can be proved rigorously. According to (16), we can write:

$$|\tilde{\psi}(t)\rangle = R(t)|\psi(t)\rangle \tag{21}$$

where $R(t)$ is the unitary operator defined by:

$$R(t) = e^{i\omega t S_z/\hbar} \tag{22}$$

We shall see later (*cf.* complement B_{VI}) that $R(t)$ describes a rotation of the coordinate system through an angle ωt about Oz. (18) is therefore indeed the transformed Schrodinger equation in the rotating $OXYZ$ frame.

Equation (18) is very simple to solve. To determine $|\tilde{\psi}(t)\rangle$, given $|\tilde{\psi}(0)\rangle$, all we need to do is expand $|\tilde{\psi}(0)\rangle$ on the eigenvectors of $\tilde{H}$ (which can be calculated exactly) and then apply rule (D-54) of chapter III (which is possible since $\tilde{H}$ is not explicitly time-dependent). We then go from $|\tilde{\psi}(t)\rangle$ to $|\psi(t)\rangle$ by using formulas (16).

C. TRANSITION PROBABILITY; RABI's FORMULA

Consider a spin which, at time $t = 0$, is in the state $|+\rangle$:

$$|\psi(0)\rangle = |+\rangle \tag{23}$$

According to (16), this corresponds to :

$$|\tilde{\psi}(0)\rangle = |+\rangle \tag{24}$$

What is the probability $\mathcal{P}_{+-}(t)$ of finding this spin in the state $|-\rangle$ at time t ? Since $a_-(t)$ and $b_-(t)$ have the same modulus, we can write :

$$\mathcal{P}_{+-}(t) = |\langle - | \psi(t) \rangle|^2 = |a_-(t)|^2 = |b_-(t)|^2 = |\langle - | \tilde{\psi}(t) \rangle|^2 \tag{25}$$

We must therefore calculate $|\langle - | \tilde{\psi}(t) \rangle|^2$, where $|\tilde{\psi}(t)\rangle$ is the solution of (18) which corresponds to the initial condition (24).

The problem we have just posed has already been solved, in § C-3-b of chapter IV. To use the calculations of that section, all we need to do is apply the following correspondences :

$$\begin{aligned} |\varphi_1\rangle &\longrightarrow |+\rangle \\ |\varphi_2\rangle &\longrightarrow |-\rangle \\ E_1 &\longrightarrow -\frac{\hbar}{2}\Delta\omega \\ E_2 &\longrightarrow \frac{\hbar}{2}\Delta\omega \\ W_{12} &\longrightarrow \frac{\hbar}{2}\omega_1 \end{aligned} \tag{26}$$

Rabi's formula [equation (C-32) of chapter IV] then becomes :

$$\mathcal{P}_{+-}(t) = \frac{\omega_1^2}{\omega_1^2 + (\Delta\omega)^2} \sin^2\left[\sqrt{\omega_1^2 + (\Delta\omega)^2}\,\frac{t}{2}\right] \tag{27}$$

The probability $\mathcal{P}_{+-}(t)$ is, of course, zero at time $t = 0$ and then varies sinusoidally with respect to time between the values 0 and $\frac{\omega_1^2}{\omega_1^2 + (\Delta\omega)^2}$. Again, we have a resonance phenomenon. For $|\Delta\omega| \gg |\omega_1|$, $\mathcal{P}_{+-}(t)$ remains almost zero (cf. fig. 3-a); near resonance, the oscillation amplitude of $\mathcal{P}_{+-}(t)$ becomes large and, when the condition $\Delta\omega = 0$ is exactly satisfied, we have $\mathcal{P}_{+-}(t) = 1$ at times $t = \frac{(2n+1)\pi}{\omega_1}$ (*cf.* fig. 3-b).

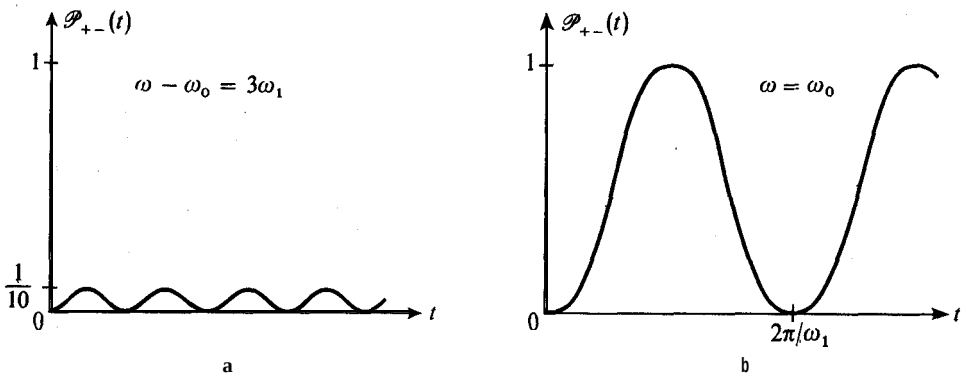


FIGURE 3

Variation with respect to time of the transition probability between the states $|+\rangle$ and $|-\rangle$, under the effect of a rotating magnetic field $B_1(t)$. Outside resonance (fig. a), this probability remains small; at resonance (fig. b), however small the field B_1, there exist times when the transition probability is equal to 1.

Thus we again find the result which we have already obtained classically : at resonance, a very weak rotating field is able to reverse the direction of the spin. Note, moreover, that the angular frequency of the oscillation of $\mathscr{P}_{+\,-}(t)$ is $\sqrt{\omega_1^2 + (\Delta\omega)^2} = |\gamma \mathbf{B}_{eff}|$. This oscillation corresponds, in the rotating frame, to the projection onto 0.2 of the precession of the magnetic moment about the effective field, sometimes called "Rabi precession" [see also the calculation of $\mathscr{P}_{+\,-}$ (t) in complement C_{IV}, § 3-c].

d. CASE WHERE THE TWO LEVELS ARE UNSTABLE

We are now going to assume that the two states $|\pm\rangle$ correspond to two sublevels of an excited atomic level (whose angular momentum is assumed equal to 1/2). n atoms are excited per unit time, all being raised to the state $|+\rangle$*. An atom decays, by spontaneous emission of radiation, with a probability per unit time of $1/\tau$, which is the same for the two sublevels $|\pm\rangle$. We know that, under these conditions, an atom which was excited at time $-t$ has a probability $e^{-t/\tau}$ of still being excited at time $t = 0$ (*cf.* complement K_{III}).

We assume that the experiment is performed in the steady state : in the presence of the fields $\mathbf{B}_0$ and $B_1(t)$, the atoms are excited at a constant rate n into the state $|+\rangle$. After a time much longer than the lifetime τ, what is the number N of atoms which decay per unit time from the state $|-\rangle$? If an atom is excited at time $-t$, the probability of finding it in the state $|-\rangle$ at $t = 0$ is $e^{-t/\tau}\mathscr{P}_{+\,-}(t)$, where $\mathscr{P}_{+\,-}(t)$ is given by relation (27). The total number of atoms in the state $|-\rangle$ is obtained by taking the sum of atoms excited at all previous times $-t$, that is, by calculating the integral :

$$\int_0^\infty e^{-t/\tau}\,\mathscr{P}_{+\,-}(t)\,n\,dt \tag{28}$$

This calculation presents no difficulties. Multiplying the number of atoms thus obtained by their probability $1/\tau$ of decay per unit time, we obtain :

$$N = \frac{n}{2}\,\frac{\omega_1^2}{(\Delta\omega)^2 + \omega_1^2 + (1/\tau)^2} \tag{29}$$

The variation of N with respect to $\Delta\omega$ corresponds to a Lorentz curve whose half-width is :

$$L = \sqrt{\omega_1^2 + (1/\tau)^2} \tag{30}$$

In the experiment described above, let us measure, for various values of the magnetic field B_0 (that is, with ω assumed to be fixed, for various values of $\Delta\omega$), the number of atoms which decay from the level $|-\rangle$. According to (29), we obtain a resonance curve which has the shape shown in figure 4.

It is very interesting to obtain such a curve experimentally, since one can use it to determine several parameters :

– if we know ω and measure the value B_0^m of the field B_0 which corresponds to the peak of the curve, we can deduce the value of the gyromagnetic ratio γ through the relation $\gamma = -\omega/B_0^m$.

* In practice, this excitation can be produced, for example, by placing the atoms in a light beam. When the incident photons are polarized, conservation of angular momentum, in certain cases, requires that the atoms which absorb them can attain only the state $|+\rangle$ (and not the state $|-\rangle$). Similarly, by detecting the polarization of the photons re-emitted by the atoms. one can know whether the atoms fall back into the ground state from the state $|+\rangle$ or the state $|-\rangle$.

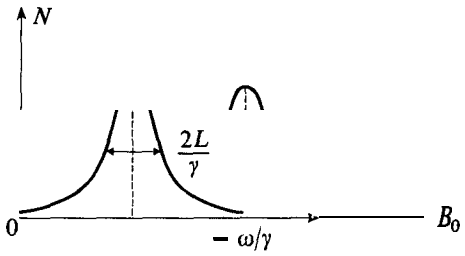


FIGURE 4

Resonance curve. To observe a resonance phenomenon, we perform an experiment in which n atoms are excited per unit time into the state $|+\rangle$. Under the effect of a field $B_1(t)$, rotating at the frequency $\omega/2\pi$, the atoms undergo transitions towards the state $|-\rangle$. In the steady state, if we measure the number N of atoms which decay per unit time from the state $|-\rangle$, we obtain a resonant variation when we scan the static field B_0 about the value $-\omega/\gamma$.

- if we know γ, we can, by measuring the frequency $\omega/2\pi$ which corresponds to resonance, measure the static magnetic field B_0. Various magnetometers, often of very great precision, operate on this principle. In certain cases, one can derive interesting information from such a measurement of the field. If, for example, the spin being considered is that of a nucleus which belongs to a molecule or to a crystal lattice, one can find the local field seen by the nucleus, its variation with the site occupied, etc.

- if we trace the square L^2 of the half-width as a function of ω_1^2, we obtain a straight line which, extrapolated to $\omega_1 = 0$, gives the lifetime τ of the excited level (*cf.* fig. 5).

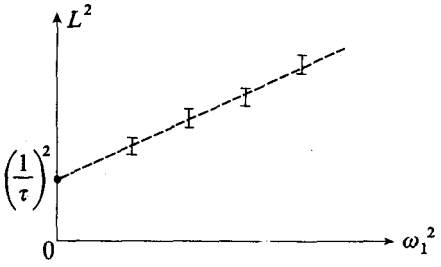


FIGURE 5

The extrapolation to $\omega_1 = 0$ of the squared half-width L of the resonance curve of figure 4 gives the lifetime of the level being studied.

3. Relation between the classical treatment and the quantum mechanical treatment: evolution of < M >

The results obtained in §§ 1 and 2 are very similar, although we used classical mechanics in one case and quantum mechanics in the other. We are now going to show that this similarity is not accidental. It arises from the fact that the quantum mechanical evolution equations of the mean value of a magnetic moment placed in an arbitrary magnetic 'field are identical to the corresponding classical equations.

The mean value of the magnetic moment associated with a spin 1/2 is:

$$\langle \mathbf{M} \rangle (t) = \gamma \langle \mathbf{S} \rangle (t) \tag{31}$$

To calculate the evolution of < M > (t), we use theorem (D-27) of chapter III :

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t} \langle \mathbf{M} \rangle (t) = \langle [\mathbf{M}, H(t)] \rangle \tag{32}$$

where H(t) is the operator:

$$\mathbf{H(t) = - M \cdot B(t)} \tag{33}$$

Let us calculate for example the commutator $[M_x, \mathrm{H(t)}]$. Using the fact that the field components $B_y(t)$ and $B_z(t)$ are numbers (*cf.* note of §2-a), we find :

$$\begin{aligned}[M_x, H(t)] &= -\gamma^2[S_x, S_xB_x(t) + S_yB_y(t) + S_zB_z(t)] \\ &= -\gamma^2 B_y(t)[S_x, S_y] - \gamma^2 B_z(t)[S_x, S_z]\end{aligned} \tag{34}$$

Using relations (14) of complement A_{IV}, we obtain :

$$[M_x, H(t)] = i\hbar\gamma^2[B_z(t)S_y - B_y(t)S_z] \tag{35}$$

Substituting (35) into (32):

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle M_x \rangle (t) = \gamma [B_z(t) \langle M_y \rangle(t) - B_y(t) \langle M_z \rangle(t)] \tag{36}$$

By cyclic permutation, we can calculate analogous expressions for the components on *Oy* and *Oz*; the three equations obtained can be condensed into:

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \mathbf{M} \rangle(t) = \gamma \langle \mathbf{M} \rangle(t) \times \mathbf{B}(t) \tag{37}$$

Let us compare (37) with (6) : the evolution of the mean value $\langle \mathbf{M} \rangle$ **(t)** obeys the classical equations exactly, whatever the time-dependence of the magnetic field **B(t).**

4. Bloch's equations

In practice, in a magnetic resonance experiment, it is not the magnetic moment of a single spin that is observed, but rather that of a great number of identical spins (as in the experiment described in §2-d above, where the number of atoms which decay from the state $|-\rangle$ is detected). Moreover, one is not concerned solely with the quantity $\mathcal{P}_{+-}(t)$, calculated above. One can also measure the global magnetization $\mathcal{M}$ of the sample under study : the sum of the mean values of the observable $\langle \mathrm{M} \rangle$ corresponding to each spin of the sample*. It is interesting, therefore, to obtain the equations of motion of $\mathcal{M}$, called the *Bloch equations.*

In order to understand the physical significance of the various terms appearing in these equations, we are going to derive them for a simple concrete case. The results obtained can be generalized to other more complicated situations.

a. A CONCRETE EXAMPLE

Consider a beam of atoms issuing from an atomic polarizer of the type studied in §B-1-a of chapter IV. All the atoms of the beam ** are in the spin state $|+\rangle$ and therefore have their magnetic moments parallel to *Oz.* They enter a cell C through a small opening (fig. 6), rebound a certain number of times from the inside walls of the cell and, after a certain time, escape through the same opening.

* It is possible to detect, for example, the electromotive force emf induced in a coil by the variation of $\mathcal{M}$ with respect to time.

** For example, silver or hydrogen atoms in the ground state. For the sake of simplicity, all effects related to nuclear spin are neglected.

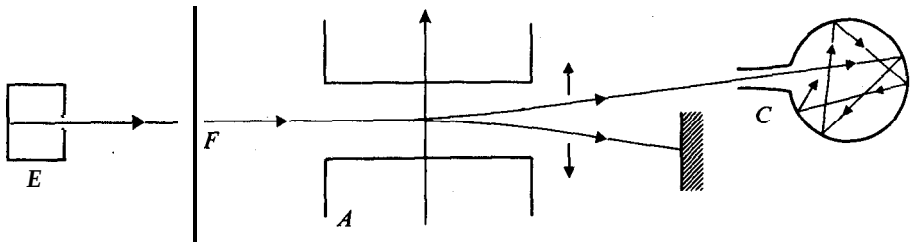


FIGURE 6

Schematic drawing of an experimental device which supplies cell C with atoms in the state $|+\rangle$.

We shall denote by n the number of polarized atoms entering the cell per unit time; n is generally small and the atomic density inside the cell is low enough to allow atomic interactions to be neglected. Moreover, if the inside walls of the cell are suitably coated, collisions with the walls have little effect on the spin state of the atoms*. We shall assume that there is a probability per unit time $1/T_R$ for the elementary magnetization introduced into the cell by a polarized atom to disappear, either because of a depolarizing collision with the walls or simply because the atom has left the cell. T_R is called the "relaxation time". The cell is placed in a magnetic field **B(t)** which may have a static component and a rotating component. The problem consists of finding the equation of motion of the global magnetization $\mathcal{M}(t)$ of the atoms which are inside the cell at time t. First, let us write the exact expression for $\mathcal{M}(t)$:

$$\mathcal{M}(t) = \sum_{i=1}^{\mathcal{N}} \langle \psi^{(i)}(t) | \mathrm{M} | \psi^{(i)}(t) \rangle = \sum_{i=1}^{\mathcal{N}} \mathcal{M}^{(i)}(t) \tag{38}$$

In (38), the sum is taken over the $\mathcal{N}$ spins which are already in the cell and which, at time t, have neither left nor undergone a depolarizing collision. $|\psi^{(i)}(t)\rangle$ is the state vector of such a spin (i) at time t [we are not counting, in (38), the spins which have undergone a depolarizing collision and have not yet left the cell, since their global contribution is zero : their spins point randomly in all directions].

Between times t and $t + \mathrm{d}t$, $\mathcal{M}(t)$ varies for three different reasons :

(i) A certain proportion, $\mathrm{d}t/T_R$, of the $\mathcal{N}$ spins undergo a depolarizing collision or leave the compartment ; these spins disappear from the sum (38) and $\mathcal{M}(t)$ therefore decreases by:

$$\mathrm{d}\mathcal{M}(t) = -\frac{\mathrm{d}t}{T_R}\mathcal{M}(t) \tag{39}$$

(ii) The other spins evolve freely in the field B(t). We saw in § 3 above that, for each of them, the evolution of the mean value of M :

$$\mathcal{M}^{(i)}(t) = \langle \psi^{(i)}(t) | \mathbf{M} | \psi^{(i)}(t) \rangle$$

obeys the classical law :

$$\mathrm{d}\mathcal{M}^{(i)}(t) = \gamma \mathcal{M}^{(i)}(t) \times \mathbf{B}(t)\, \mathrm{d}t \tag{40}$$

Since the right-hand side of (40) is linear with respect to $\mathcal{M}^{(i)}(t)$, the contribution of these spins to the variation of $\mathcal{M}(t)$ is given by :

$$\mathrm{d}\mathcal{M}(t) = \gamma \mathcal{M}(t) \times \mathrm{B}(t)\, \mathrm{d}t \tag{41}$$

* For example, for hydrogen atoms bouncing off teflon walls, tens of thousands of collisions are required for the magnetic moment of the hydrogen atom to become disoriented.

(iii) Finally, a certain number, n dt, of new spins have entered the cell. Each of them adds to the global magnetization a contribution $\boldsymbol{\mu}_0$, equal to the mean value of M in the state $|+\rangle$ ($\boldsymbol{\mu}_0$ is parallel to *Oz* and $|\boldsymbol{\mu}_0| = |\gamma| \frac{\hbar}{2}$). $\mathcal{M}$ therefore increases by :

$$dM(t) = n\,\boldsymbol{\mu}_0\,dt \tag{42}$$

The global variation of $\mathcal{M}$ is obtained by adding (39), (41) and (42). Dividing by dt, we obtain the equation of motion of $\mathcal{M}(t)$ (Bloch equation):

$$\frac{d}{dt}\mathcal{M}(t) = n\,\boldsymbol{\mu}_0 - \frac{1}{T_R}\mathcal{M}(t) + \gamma\mathcal{M}(t) \times \mathbf{B}(t) \tag{43}$$

We have derived (43) in a specific case, making certain hypotheses. However, the main features of this equation remain valid for a great number of other experiments where the rate of variation of 1(t) appears in the form of a sum of three terms :

- a source term (here $n\,\boldsymbol{\mu}_0$) which describes the preparation of the system. It would, in fact, be impossible to observe magnetic resonance without a preliminary polarization of the spins, which can be achieved through selection using a magnetic field gradient (as in the example studied here), a polarized optical excitation (as in the example studied in 52-d above), cooling of the sample in a strong static field, etc.

- a damping term ($\text{here} - \frac{1}{T_R}\mathcal{M}(t)$) which describes the disappearance or "relaxation" of the global magnetization under the effect of various processes : collisions, disappearance of atoms, change in atomic levels through spontaneous emission (as in the example studied in §2-d), etc.

- a term which describes the precession of $\mathcal{M}(t)$ in the field B(r) [last term of (43)].

b. SOLUTION IN THE CASE OF A ROTATING FIELD

When the field B(t) is the sum of a static field $\mathbf{B}_0$ and a rotating field B,(t), such as those considered above, equations (43) can be solved exactly. As in §§ 1 and 2, one changes to the rotating frame *OXYZ*, with respect to which the relative variation of $\mathcal{M}$ (t) is :

$$\left(\frac{d}{dt}\mathcal{M}\right)_{rel} = n\,\boldsymbol{\mu}_0 - \frac{1}{T_R}\mathcal{M} + \gamma\mathcal{M} \times \mathbf{B}_{eff} \tag{44}$$

[where $\mathbf{B}_{eff}$ is defined by equation (10)].

Projecting this equation onto *OX*, *OY* and *OZ*, we obtain a system of three linear differential equations with constant coefficients whose stationary solution (valid after a time much greater than T_R) is:

$$\begin{aligned}
(\mathcal{M}_X)_S &= -\,n\mu_0 T_R \frac{\omega_1 \Delta\omega}{(\Delta\omega)^2 + \omega_1^2 + (1/T_R)^2} \\
(\mathcal{M}_Y)_S &= -\,n\mu_0 \frac{\omega_1}{(\Delta\omega)^2 + \omega_1^2 + (1/T_R)^2} \\
(\mathcal{M}_Z)_S &= n\mu_0 T_R \left[1 - \frac{\omega_1^2}{(\Delta\omega)^2 + \omega_1^2 + (1/T_R)^2}\right]
\end{aligned} \tag{45}$$

The three components of the stationary magnetization $(\mathcal{M})_S$, when the field B_0 varies, have resonant variations about the value $B_0 = -\omega/\gamma$ (*cf.* fig. 7). $(\mathcal{M}_Y)_S$ and $(\mathcal{M}_Z)_S$ give absorption curves (Lorentz curves of width $\frac{2}{\gamma}\sqrt{\omega_1^2 + (1/T_R)^2}$). $(\mathcal{M}_X)_S$ gives a dispersion curve (of the same width).

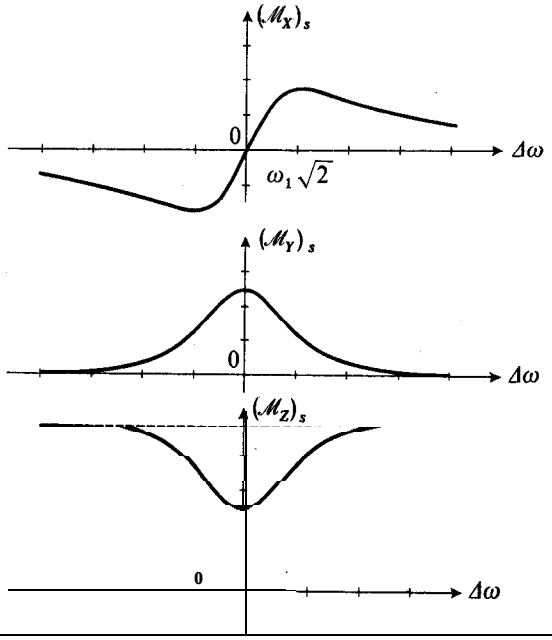


FIGURE 7

Variation with respect to $\Delta\omega = \omega - \omega_0$ of the stationary values of the components of $\mathcal{M}$ in the rotating frame. One obtains a dispersion curve for $(\mathcal{M}_X)_S$ and absorption curves for $(\mathcal{M}_Y)_S$ and $(\mathcal{M}_Z)_S$. The three curves have the same width, $2\sqrt{\omega_1^2 + (1/T_R)^2}$, which increases with ω_1. They have been drawn assuming that $\omega_1 = 1/T_R$ ("half-saturation").

The comments made at the end of 52-d above about the experimental interest of such curves can be repeated here.

References and suggestions for further reading :

Feynman II (7.2), chap. 35; Cagnac and Pebay-Peyroula (11.2), chaps. IX §5, X S5, XI §§2 to 5, XIX S3; Kuhn (11.1), § VI, D.

See the references of section 14 of the bibliography, particularly Abragam (14.1) and Slichter (14.2).